

D.M. RAFIUN BIN MASUD (AI Engineer | Machine Learning Engineer)

AI Engineer specializing in machine learning, deep learning, and computer vision with experience developing and deploying AI applications using Python, TensorFlow, Scikit-learn, OpenCV, FastAPI, Docker, and Git. Experienced in the complete ML lifecycle, from data preprocessing and model training to API development, containerized deployment, and production support.



☎ 01670373934

✉ dmrafiun@gmail.com

in [linkedin.com/in/dmrafiun](https://www.linkedin.com/in/dmrafiun)

🐙 github.com/arnab9961

📍 Dhaka

EDUCATIONAL

B.Sc in Computer Science & Engineering (CSE)

East West University (EWU)

2024 | CGPA 3.05/4.00

WORK EXPERIENCE

AI Engineer

Betopia | Dhaka, Bangladesh

Mar 2025 – Present

- Led a team of junior AI engineers in developing and deploying production-ready ML solutions.
- Developed and deployed AI solutions for recommendation systems, computer vision, and speech analysis across multiple production projects.
- Managed the end-to-end ML lifecycle, from data preprocessing and model training to Docker-based deployment.
- Collaborated with product managers and cross-functional teams to deliver scalable AI features.
- Oversaw industrial ML projects including:
 - Dating App Matching & Filtering Algorithms.
 - Quran Pronunciation Evaluation System (Audio Analysis).
 - Food Calorie Estimation via Image Processing.

IT Support Engineer

Taltola Dot Net | Dhaka, Bangladesh

Mar 2021 – Jan 2022

- Optimized IT infrastructure and managed network configurations through installations, maintenance, and upgrades, ensuring reliable performance with minimal disruptions.
- Enhanced security by improving protection protocols, monitoring system performance, resolving bottlenecks, and training employees on effective IT system usage.

CORE COMPETENCIES

Software Development • Data Analysis • Machine Learning • Automation Testing • System Optimization • Data Preprocessing • Predictive Modeling • Model Training • Model Evaluation • Deep Learning • Data Visualization • Regression Analysis • Feature Extraction • QA Automation • Performance Monitoring • Algorithm Development • Statistical Analysis • Testing Frameworks • Data Mining • Problem Solving • Process Automation

TECHNICAL SKILLS

Programming: Python, JavaScript, C, C++ • **AI/ML:** TensorFlow, Keras, Scikit-learn, OpenCV, Pandas, NumPy • **Backend & APIs:** FastAPI, REST APIs • **DevOps:** Docker, Git, GitHub, VPS Deployment • **Databases:** MySQL, Oracle SQL • **Web:** React, HTML, CSS, PHP • **Tools:** VS Code, PyCharm, Postman

LANGUAGES

Bengali (Native) | **English** (Fluent)

PROJECT WORK

Ball Shundari Apple Plum Growth Stage Classification

- Collected and processed a dataset of 701 images of Ball Shundari apple plums.
- Applied advanced image processing techniques, including resizing, normalization, and augmentation.

MANIFEX_AI: AI-powered paragraph evaluation platform using NLP to assess sentence structure, grammar, vocabulary, and writing quality.

Tech Stack: Python, FastAPI, NLP, Docker

LAMARE: MedGemma-Based Smart Healthcare Platform for Wound Analysis , Surgical Outcome Simulation and NHS Injectable Medicines.

Tech Stack: MedGemma, Python, FastAPI

DFT_Image_Prep: Upscayl Enhanced Image and Performs image enhancement, background removal, and automatic trimming using advanced image processing algorithms. AI-based image preprocessing tool for image enhancement, background removal, automatic trimming, and quality improvement.

Research Publications & Projects

Deep Learning and Non-Contrast CT for Widespread Pancreatic Cancer Detection

Best Paper Award – International Conference on Innovations in Data Analytics (2022)

- Investigated the application of deep learning models for early pancreatic cancer detection using non-contrast CT scans, designing and training convolutional neural networks (CNNs) to identify cancerous tissues.
- Preprocessed large CT image datasets to improve model performance and reliability, and evaluated outcomes to achieve higher precision in identifying early-stage pancreatic cancer.

Academic Project: Machine Learning-Based Classification of Slow Growth Stages in the Ball Shundari Apple Plum Image Dataset (2022)

- Developed a machine learning model to classify the growth stages of Ball Shundari apple plums using image datasets, applying feature extraction techniques to capture key image characteristics.
- Trained, tested, and optimized multiple classification algorithms for accuracy and efficiency, validating model predictions through statistical analysis against ground truth data.

SFiNNet: Leveraging Transfer Learning and Explainable AI for Automated Small Fish Classification Enhanced via Adversarial Training and Augmented Regularization

- Proposed an intelligent fish classification framework using transfer learning, explainable AI, adversarial training, and augmented regularization.
- Improved classification performance while delivering transparent, reliable, and interpretable predictions for small fish species identification.

Certifications

Accenture North America – Data Analytics and Visualization Job Simulation – 2024 | Forage

EXTRACURRICULAR ACTIVITIES

Event Coordinator | IEEE Computer Society Student Branch